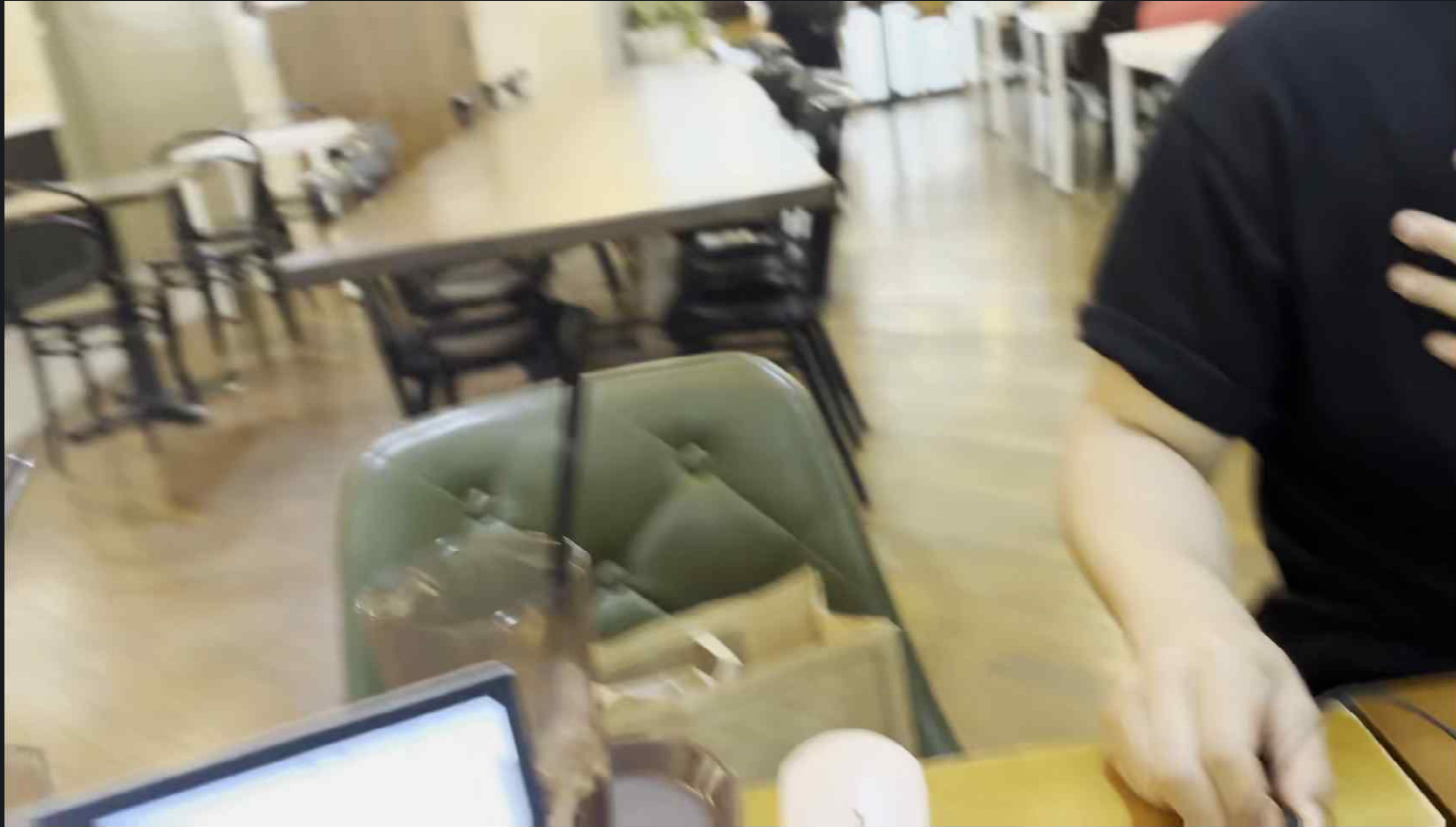
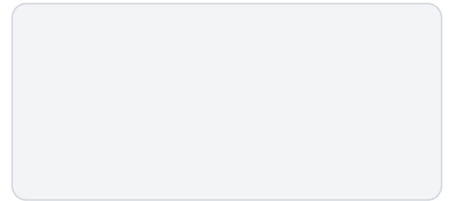
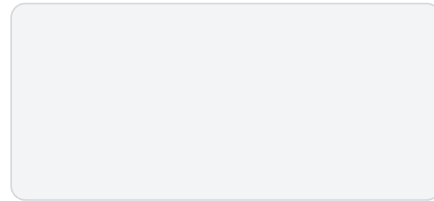
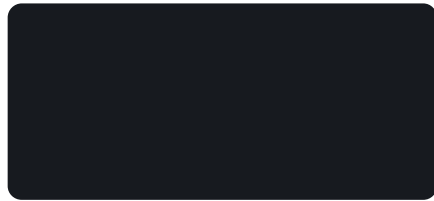
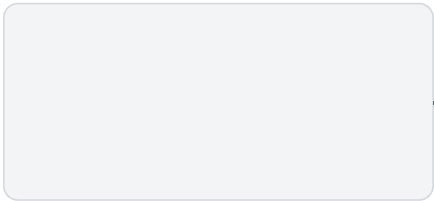
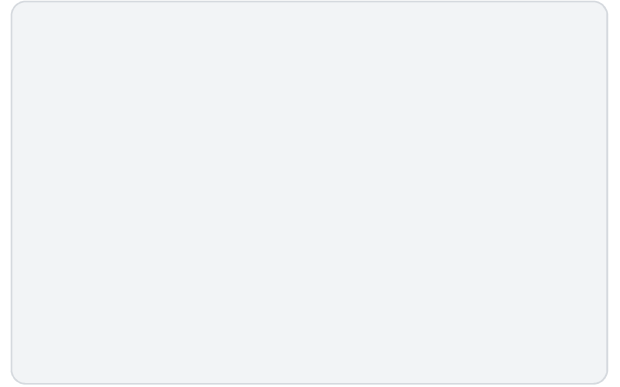
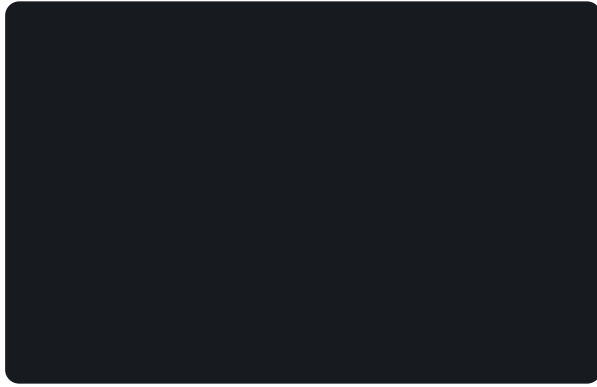
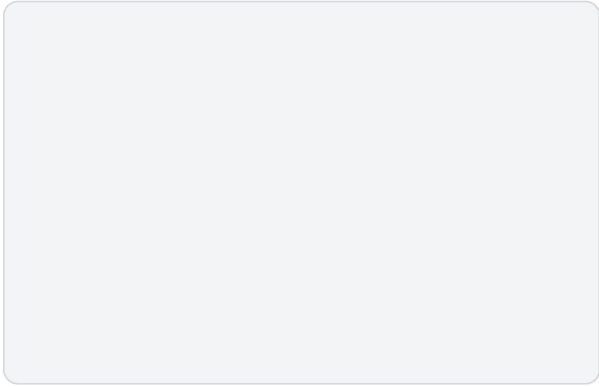


Edge AI Threat-Sound Detection



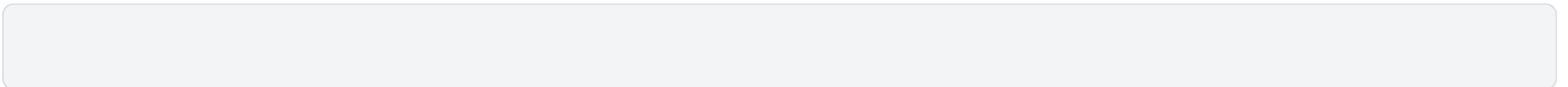
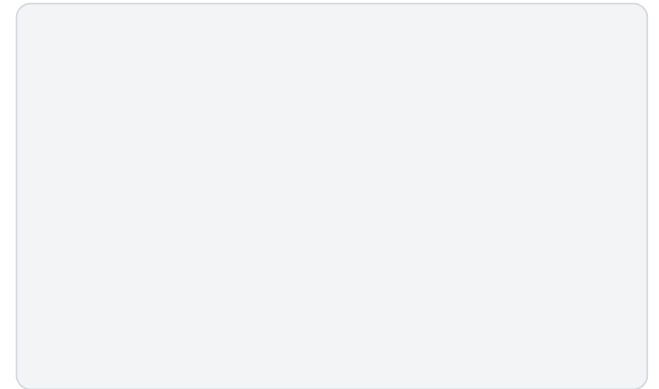
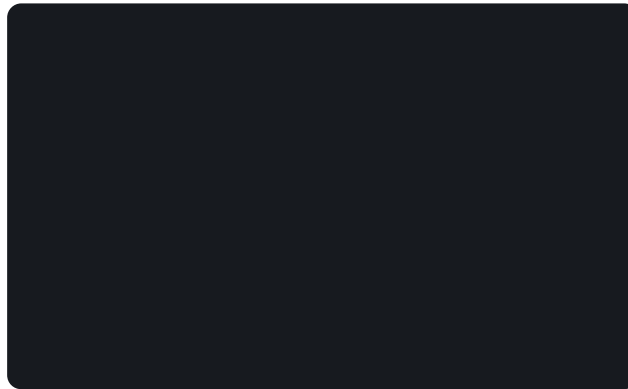
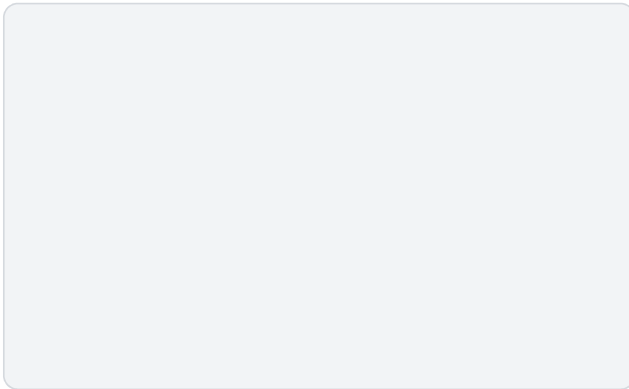
Demo Video

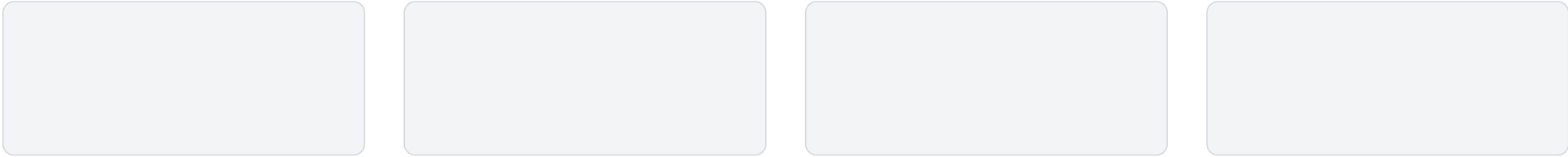
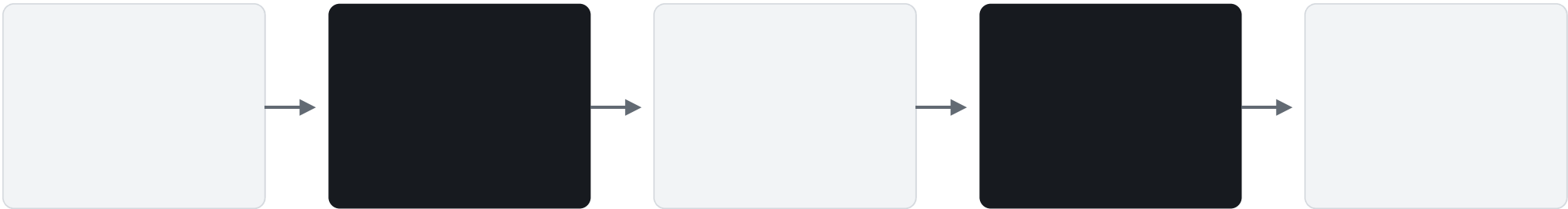


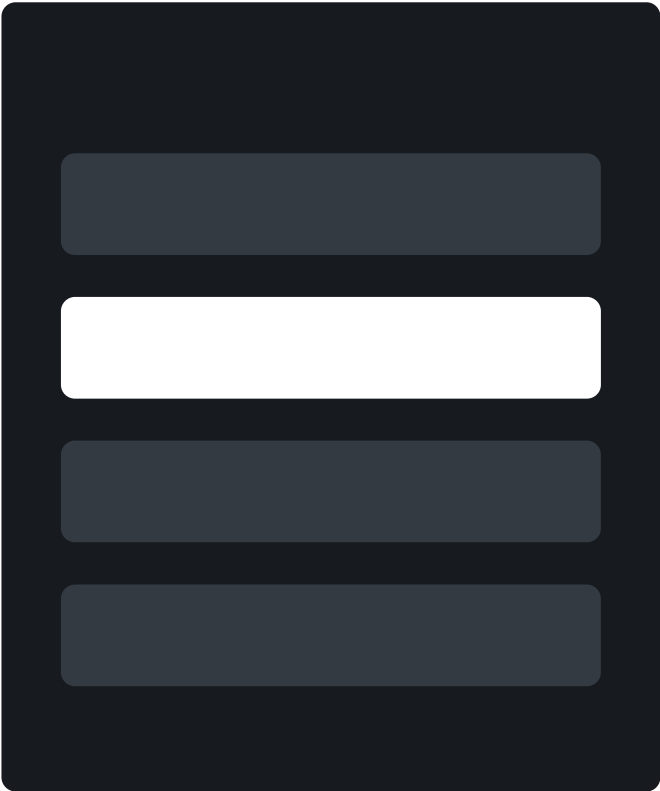


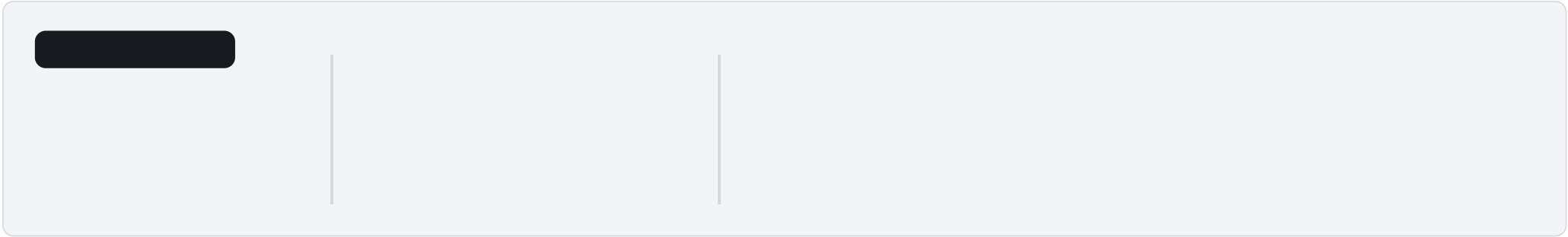
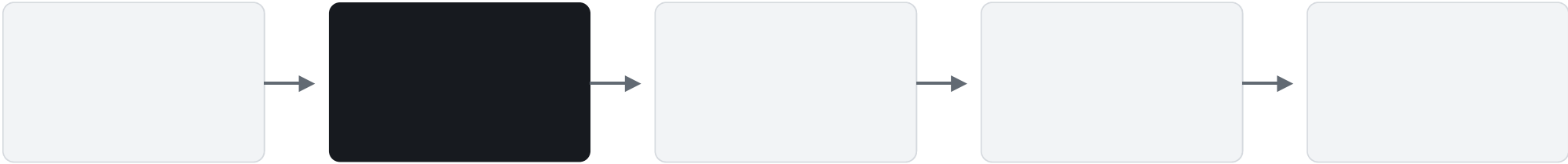
MAIN CONCEPT

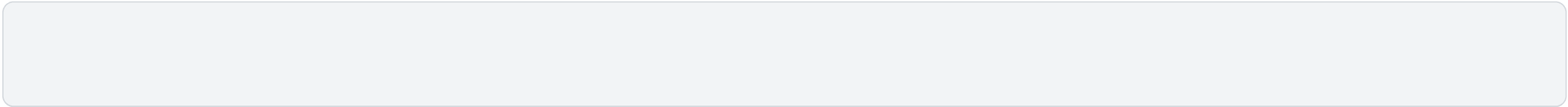
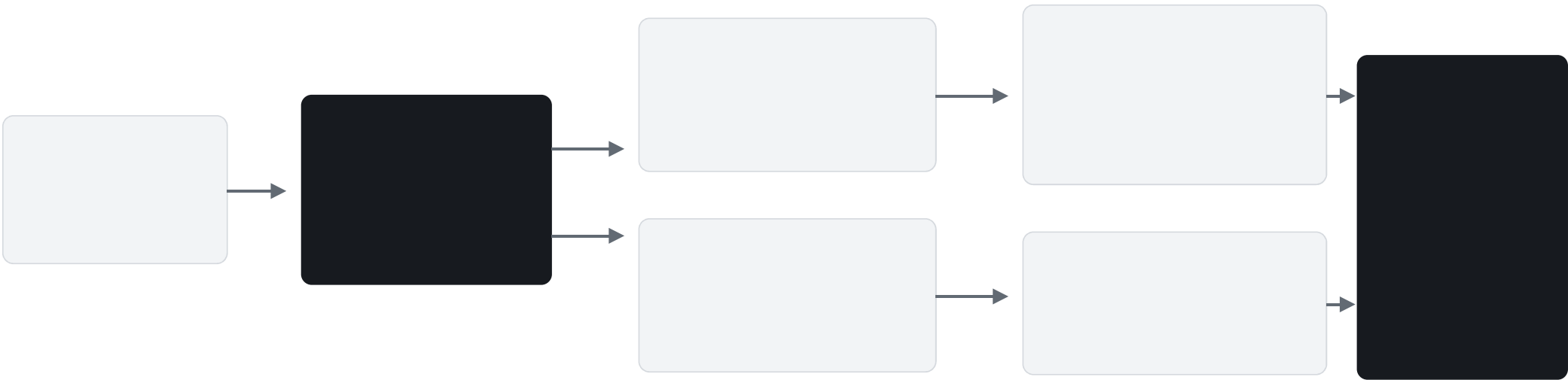
Use sound as an additional security sensor, classify threats locally, and connect detection directly to IoT response.

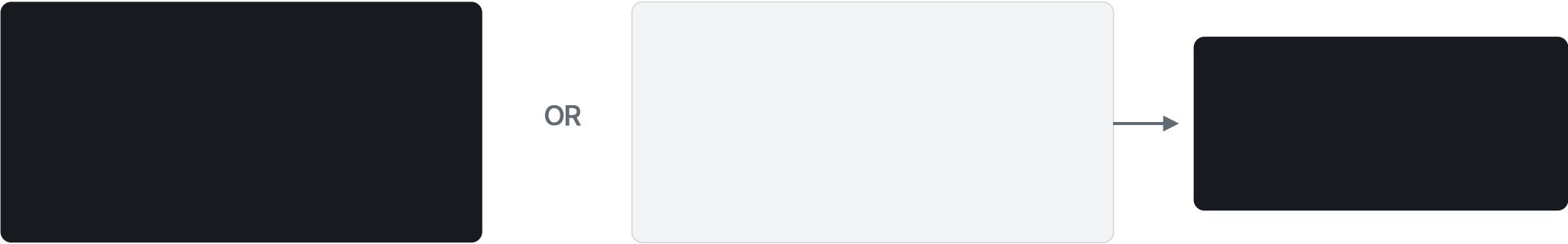












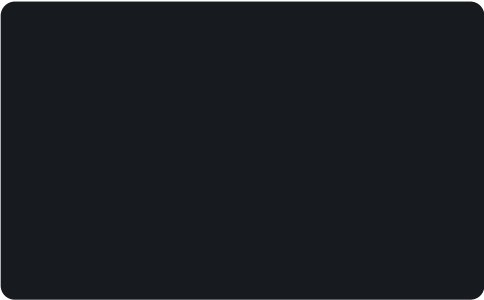
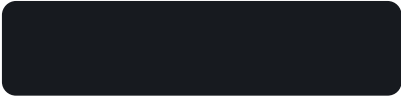
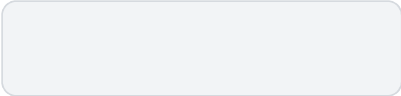
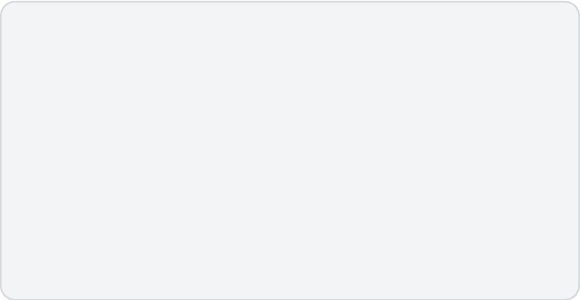
Class	Strong custom	Strong frames	Aligned custom	YAMNet score	Aligned frames

Screaming · Glass · Shatter

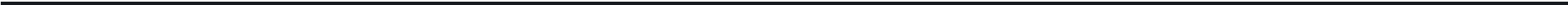
reviewed failures · live examples

speech · crowds · music · impacts ·
footsteps

16 kHz mono · 3-second clips · strict target labels · original-source identity retained



Total	732	4,458	5,190	156	165



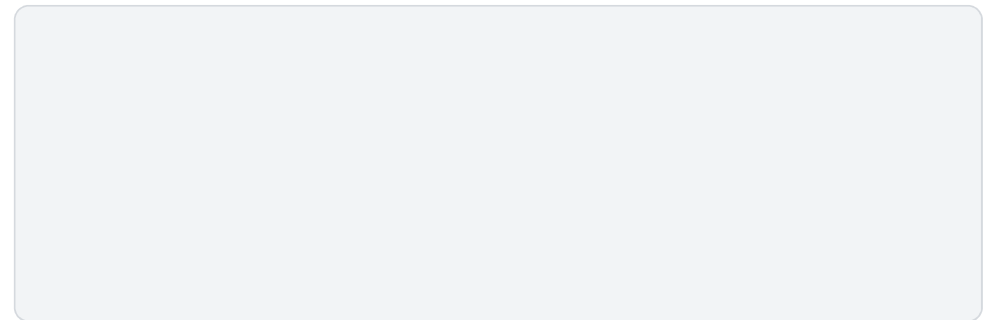
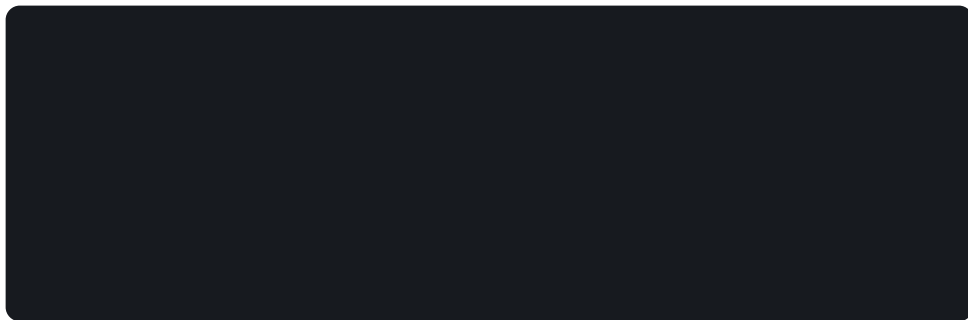
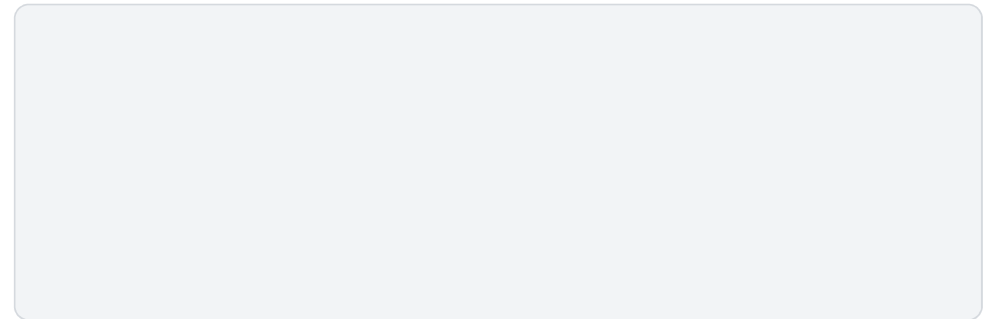
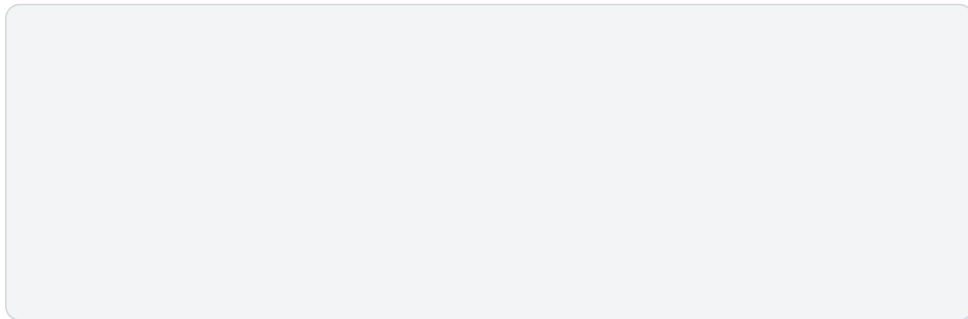
1 

2 

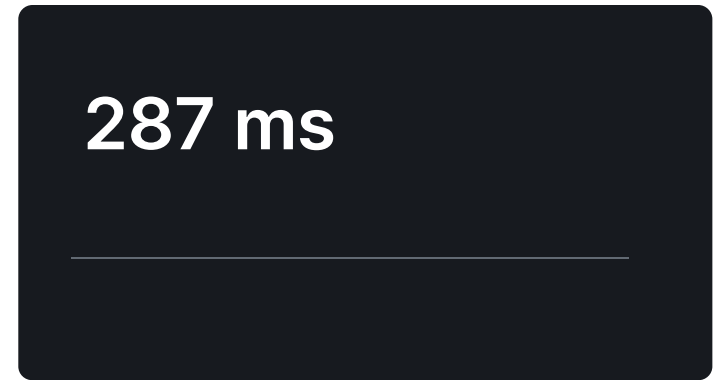
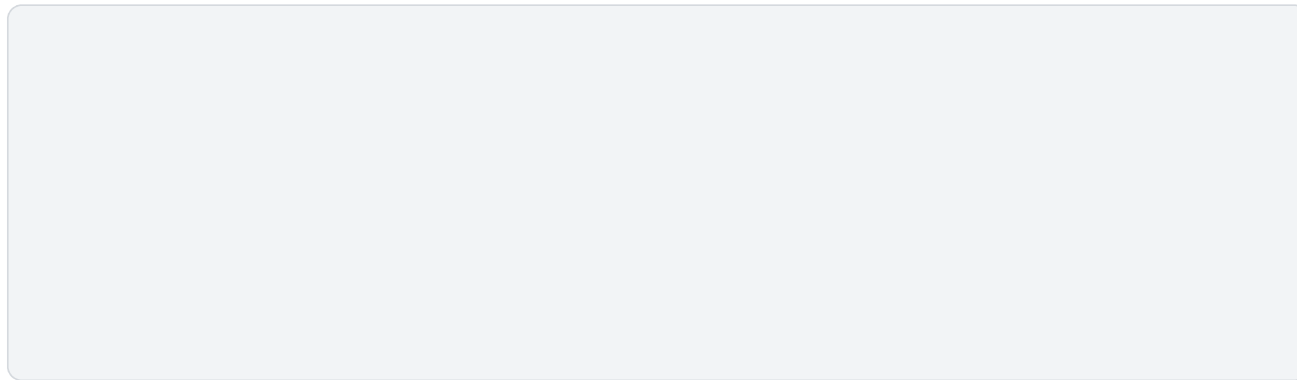
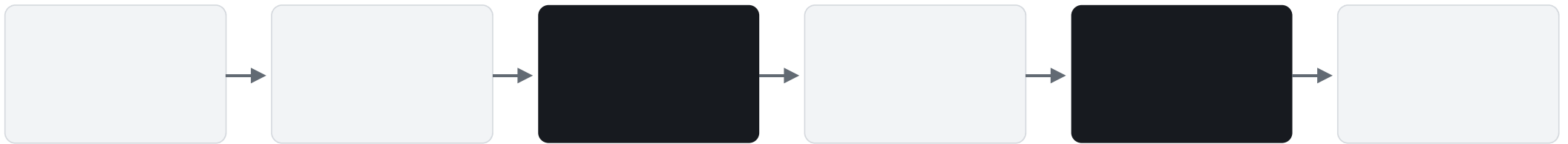
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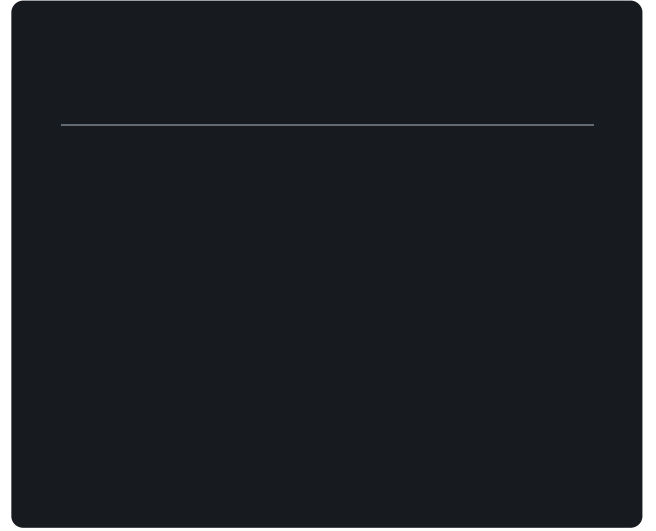
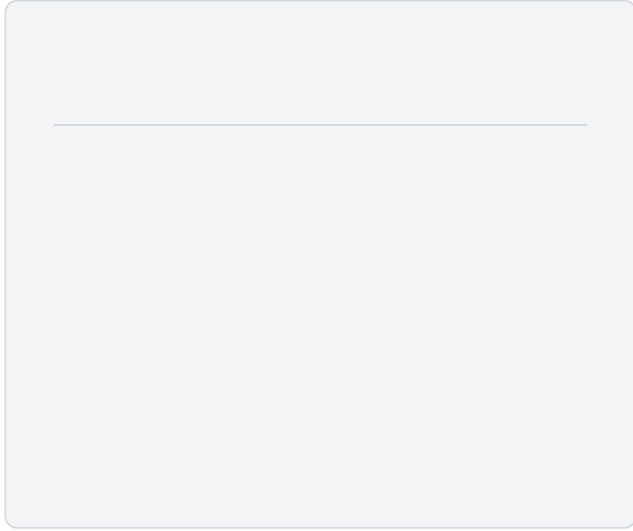
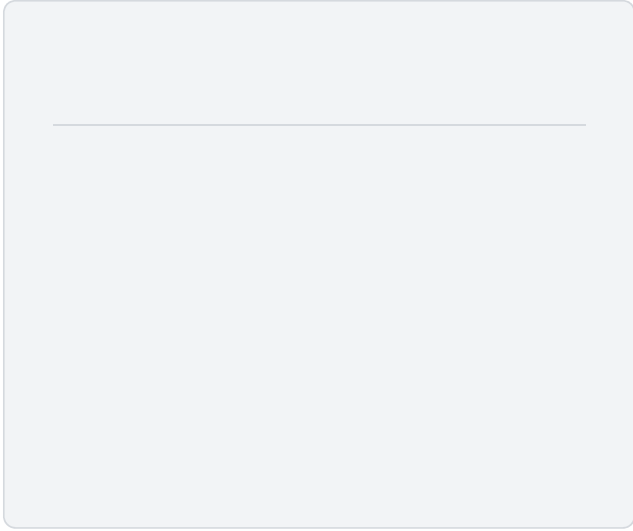
4 

5 



BENEFIT · Local inference reduces continuous audio transmission and supports immediate physical and remote response.





Security that listens.
